

Chapter 7: Calculus of Variations

Mathematical Physics

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Outline

- 1 From a functional to an equation
- 2 Energy and complex analysis
- 3 Normalization and Sturm–Liouville modes
- 4 Disk modes and a computable approximation
- 5 Other special functions and the complete picture

One question connects the course

Earlier chapters started with an equation and found its solutions. Here we ask:

Why does that equation arise, and what does its solution optimize?

The common route is

energy of a function $\longrightarrow$ small change of the function $\longrightarrow$ differential equation.

- With fixed boundary values: Laplace's equation and analytic functions.
- With fixed size: Sturm–Liouville eigenfunctions, including Bessel and polynomial families.
- With a trial shape: an approximation whose integrals use beta and gamma functions.

We use real-valued trial functions. Complex functions remain a tool for constructing real harmonic fields. Differentiation, integration by parts, and the earlier chapters supply the calculations.

A functional assigns a number to a whole curve

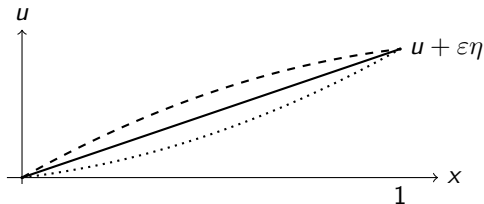
An ordinary function takes a number as input. A *functional* takes a function:

$$u(x) \longmapsto J[u] = \frac{1}{2} \int_0^1 u'(x)^2 dx.$$

This energy penalizes rapid change. Suppose $u(0) = 0$ and $u(1) = 1$ are fixed. To change the curve without moving its ends, write

$$u_\varepsilon(x) = u(x) + \varepsilon\eta(x), \quad \eta(0) = \eta(1) = 0. \quad (7.1.1)$$

Here η chooses the *shape* of the change; the real number ε chooses its size and sign.



The first variation is an ordinary derivative

Keep u and η fixed and differentiate with respect to ε , not x . For the preceding energy the expansion is exact:

$$\begin{aligned} J[u + \varepsilon\eta] &= \frac{1}{2} \int_0^1 (u' + \varepsilon\eta')^2 dx \\ &= J[u] + \varepsilon \int_0^1 u'\eta' dx + \frac{\varepsilon^2}{2} \int_0^1 \eta'^2 dx. \end{aligned}$$

The coefficient of ε is the *first variation*:

$$\delta J[u; \eta] := \left. \frac{d}{d\varepsilon} J[u + \varepsilon\eta] \right|_{\varepsilon=0} = \int_0^1 u'\eta' dx. \quad (7.1.2)$$

At a minimum it must vanish for **every allowed** η . Otherwise, changing the sign of a sufficiently small ε would lower J . A function with zero first variation is called *stationary*.

One integration by parts gives the equation

Assume u is twice continuously differentiable. Then

$$\delta J = [u'\eta]_0^1 - \int_0^1 u''\eta \, dx = - \int_0^1 u''\eta \, dx,$$

since the endpoint values of η are zero. Why does this imply $u'' = 0$? If u'' were positive or negative somewhere, continuity would give a small interval with that sign. A nonnegative smooth bump η inside that interval would make the integral nonzero. Thus

$$u'' = 0 \quad \implies \quad u(x) = A + Bx.$$

The endpoint data give $A = 0$, $B = 1$, so $u_*(x) = x$. For *any* smooth η with zero endpoints,

$$J[u_* + \eta] - J[u_*] = \frac{1}{2} \int_0^1 \eta'^2 \, dx \geq 0. \quad (7.1.3)$$

Equality forces $\eta' = 0$ and then $\eta = 0$. This proves a unique minimum; stationarity alone would not. For example, $-J$ has a maximum at the same curve.

The general Euler–Lagrange calculation

Let

$$J[u] = \int_a^b F(x, u(x), u'(x)) \, dx.$$

View $F(x, s, v)$ as a function of three independent inputs. F_s differentiates with respect to the value, and F_v with respect to the slope. For smooth F, u, η , the chain rule gives

$$\begin{aligned} \delta J[u; \eta] &= \int_a^b (F_s \eta + F_v \eta') \, dx \\ &= [F_v \eta]_a^b + \int_a^b \left(F_s - \frac{d}{dx} F_v \right) \eta \, dx. \end{aligned}$$

All derivatives of F are evaluated at $(x, u(x), u'(x))$. Using variations that vanish at the ends gives the *Euler–Lagrange equation*:

$$\frac{d}{dx} F_v = F_s. \tag{7.1.4}$$

The endpoint term determines the boundary condition

After the Euler–Lagrange equation holds, the first variation is

$$\delta J = F_v(b)\eta(b) - F_v(a)\eta(a).$$

Fixed endpoint value: $u(b)$ is prescribed, so $\eta(b) = 0$. There is no extra equation for $u'(b)$.

Free endpoint value: the location b is fixed, but $u(b)$ is not prescribed. Then $\eta(b)$ is arbitrary, so stationarity requires $F_v(b) = 0$. This is a *natural boundary condition*.

For $F = \frac{1}{2}p(x)u'^2$, it reads $p(b)u'(b) = 0$. If $p(b) > 0$, this is $u'(b) = 0$. If p vanishes or an endpoint is infinite, the limiting boundary term must be checked rather than divided by p .

The equation and the boundary conditions come from the same calculation. We do not vary the endpoint locations in this chapter.

A complete example with one free endpoint

Consider

$$J[u] = \int_0^1 \left(\frac{1}{2} u'^2 - u \right) dx, \quad u(0) = 0, \quad u(1) \text{ free.}$$

Here $F_s = -1$, $F_v = u'$, so

$$u'' = -1, \quad u'(1) = 0.$$

Integrate twice and use the conditions:

$$u' = -x + C, \quad C = 1; \quad u = -\frac{x^2}{2} + x + D, \quad D = 0.$$

For an allowed perturbation $\eta(0) = 0$, with $\eta(1)$ unrestricted,

$$\begin{aligned} J[u + \eta] - J[u] &= \int_0^1 (u' \eta' - \eta) dx + \frac{1}{2} \int_0^1 \eta'^2 dx \\ &= [u' \eta]_0^1 - \int_0^1 (u'' + 1) \eta dx + \frac{1}{2} \int_0^1 \eta'^2 dx = \frac{1}{2} \int_0^1 \eta'^2 dx. \end{aligned}$$

Thus this stationary solution is also the unique minimum.

The same energy in the plane

Let Ω be a bounded connected region with a piecewise smooth boundary. For a real field $u(x, y)$, define the *Dirichlet energy*

$$\mathcal{D}[u] = \frac{1}{2} \iint_{\Omega} (u_x^2 + u_y^2) \, dA = \frac{1}{2} \iint_{\Omega} |\nabla u|^2 \, dA. \quad (7.2.1)$$

Here $\nabla u = (u_x, u_y)$, $dA = dx \, dy$, and u_x means differentiate in x while holding y fixed. The integral adds local energy over area. Prescribe $u = g$ on the boundary; hence $\eta = 0$ there.

Expand both squares just as in one dimension:

$$\delta \mathcal{D}[u; \eta] = \iint_{\Omega} (u_x \eta_x + u_y \eta_y) \, dA.$$

The next step is still integration by parts, now in the two coordinate directions. We use smooth fields and finite integrals throughout this calculation.

Planar integration by parts produces Laplace's equation

The product rule gives

$$u_x \eta_x + u_y \eta_y = \partial_x(\eta u_x) + \partial_y(\eta u_y) - \eta(u_{xx} + u_{yy}).$$

The planar divergence theorem turns the first two terms into a boundary integral:

$$\delta \mathcal{D} = \int_{\partial \Omega} \eta \partial_n u \, ds - \iint_{\Omega} \eta \Delta u \, dA. \quad (7.2.2)$$

Here $\mathbf{n}$ is the outward unit normal, $\partial_n u = \nabla u \cdot \mathbf{n}$, ds is boundary length, and $\Delta u = u_{xx} + u_{yy}$ is the Laplacian. This boundary identity is the two-dimensional integration-by-parts rule.

Since $\eta = 0$ on the boundary, interior bump variations imply

$$\Delta u = 0 \quad \text{in } \Omega. \quad (7.2.3)$$

Thus a stationary field is *harmonic*, the class encountered in Chapter 6.

Why the harmonic solution is a minimum

Suppose a smooth harmonic function u_* with the prescribed boundary values has been found. Any competitor is $u_* + \eta$, where $\eta = 0$ on the boundary. The exact square expansion gives

$$\begin{aligned}\mathcal{D}[u_* + \eta] - \mathcal{D}[u_*] &= \iint_{\Omega} \nabla u_* \cdot \nabla \eta \, dA + \frac{1}{2} \iint_{\Omega} |\nabla \eta|^2 \, dA \\ &= \frac{1}{2} \iint_{\Omega} |\nabla \eta|^2 \, dA \geq 0.\end{aligned}$$

The cross term vanishes by (7.2.2), since $\Delta u_* = 0$. Equality means $\nabla \eta = 0$. On a connected region, η is then constant; its zero boundary values force that constant to be zero.

Therefore the harmonic solution is the unique energy minimizer. This proves minimality and uniqueness once a solution is constructed; it does not by itself prove existence for arbitrary boundary data.

Analytic functions construct the minimizers

Return to Chapter 1. If $F(z) = u(x, y) + iv(x, y)$ is analytic, the Cauchy–Riemann equations are

$$u_x = v_y, \quad u_y = -v_x.$$

Differentiate the first in x and the second in y :

$$u_{xx} = v_{yx}, \quad u_{yy} = -v_{xy}, \quad \Delta u = v_{yx} - v_{xy} = 0.$$

The mixed derivatives agree because an analytic function is smooth. Hence $\operatorname{Re} F$ minimizes Dirichlet energy for its own boundary data.

For example, $F(z) = z^n$ gives

$$u(r, \theta) = r^n \cos(n\theta), \quad n = 0, 1, 2, \dots,$$

using $z = re^{i\theta}$. These are precisely the regular disk harmonics from Chapter 6. The same solution now has both an *analytic* and a *variational* description.

From boundary Fourier data to an analytic polynomial

On the unit circle, prescribe a finite trigonometric sum with real coefficients:

$$g(\theta) = a_0 + \sum_{n=1}^M (a_n \cos n\theta + b_n \sin n\theta).$$

Choose

$$F(z) = a_0 + \sum_{n=1}^M (a_n - ib_n)z^n.$$

Indeed, $\operatorname{Re}[(a_n - ib_n)e^{in\theta}] = a_n \cos n\theta + b_n \sin n\theta$. Therefore

$$u_*(r, \theta) = a_0 + \sum_{n=1}^M r^n (a_n \cos n\theta + b_n \sin n\theta) \quad (7.2.4)$$

fits the boundary and is the unique energy minimizer. As in Chapter 1, integrate counterclockwise to extract its coefficients:

$$a_n - ib_n = \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{F(\zeta)}{\zeta^{n+1}} d\zeta, \quad n = 1, \dots, M.$$

A worked disk boundary-value problem

Let $u(1, \theta) = 2 + 3 \cos \theta - \sin 2\theta$. The preceding construction gives

$$F(z) = 2 + 3z + iz^2, \quad u_* = \operatorname{Re} F = 2 + 3x - 2xy.$$

This is harmonic because $u_{*,xx} = u_{*,yy} = 0$. On $x = \cos \theta$, $y = \sin \theta$ it has the required boundary values. Its energy can also be calculated:

$$\begin{aligned} 2\mathcal{D}[u_*] &= \iint_{x^2+y^2 < 1} ((3-2y)^2 + 4x^2) \, dA \\ &= 9\pi + 4 \iint_{x^2+y^2 < 1} (x^2 + y^2) \, dA \\ &= 9\pi + 4 \int_0^{2\pi} \int_0^1 r^3 \, dr \, d\theta = 11\pi. \end{aligned}$$

The term linear in y integrates to zero by reflection symmetry. Thus $\mathcal{D}[u_*] = 11\pi/2$, and every different smooth field with the same boundary values has larger energy.

An annulus brings back the logarithm

Let $a < r < b$, with $0 < a < b$, and prescribe $u(a) = 0$, $u(b) = 1$ on the two circles. Try a radial field. Since $dA = r dr d\theta$ and $|\nabla u|^2 = u'(r)^2$,

$$\mathcal{D}[u] = \pi \int_a^b r u'(r)^2 dr.$$

The Euler–Lagrange equation for $F = \pi r u'^2$ is

$$(2\pi r u')' = 0 \implies r u' = C, \quad u = C \log r + D.$$

The two boundary values give

$$u_*(r) = \frac{\log(r/a)}{\log(b/a)}. \tag{7.2.5}$$

Because $\Delta u_* = (r u'_*)'/r = 0$ on the entire annulus, the planar minimum argument applies to *all* smooth competitors, not merely radial ones.

The field is global; its analytic potential need not be

The annular solution is single-valued because $r = |z|$ is single-valued. Locally it is the real part of

$$\frac{\log z - \log a}{\log(b/a)}.$$

Its local imaginary part (harmonic conjugate) is $\theta/\log(b/a)$. One turn around the hole adds $2\pi/\log(b/a)$ to that conjugate. Thus there is **no single-valued analytic potential on the whole annulus** with this real part. This is the branch issue from Chapter 1, not a failure of harmonicity.

The energy is nevertheless unambiguous:

$$\mathcal{D}[u_*] = \frac{\pi}{\log^2(b/a)} \int_a^b \frac{dr}{r} = \frac{\pi}{\log(b/a)}.$$

A hole changes the possible analytic potentials; it does not invalidate the variational problem with fixed boundary values.

Analytic coordinate changes: the gradient calculation

Let $w = f(z) = X(x, y) + iY(x, y)$ map Ω one-to-one onto $\tilde{\Omega}$, with f analytic and $f' \neq 0$. Such a map is *conformal*. Write $f' = a + ib$. Cauchy–Riemann gives

$$X_x = Y_y = a, \quad Y_x = -X_y = b.$$

For a field $U(X, Y)$, set $u(x, y) = U(X(x, y), Y(x, y))$. The ordinary chain rule yields

$$u_x = aU_X + bU_Y, \quad u_y = -bU_X + aU_Y.$$

Squaring and adding cancels the cross terms:

$$|\nabla u|^2 = (a^2 + b^2)(U_X^2 + U_Y^2) = |f'|^2 |\nabla U|^2. \quad (7.2.6)$$

The map locally rotates and scales both coordinate directions by the same factor. The next page shows why that scaling cancels in the energy.

Why planar Dirichlet energy is conformally invariant

The area scale is the determinant

$$\frac{\partial(X, Y)}{\partial(x, y)} = \begin{vmatrix} a & -b \\ b & a \end{vmatrix} = a^2 + b^2 = |f'|^2.$$

Thus $dA_w = |f'|^2 dA_z$, so (7.2.6) gives

$$\mathcal{D}_\Omega[U \circ f] = \frac{1}{2} \iint_\Omega |f'|^2 |\nabla U|^2 dA_z = \frac{1}{2} \iint_{\tilde{\Omega}} |\nabla U|^2 dA_w. \quad (7.2.7)$$

For maps that also give a smooth one-to-one correspondence of the boundaries, corresponding boundary data have corresponding energy minimizers.

A useful limit on this statement: the squared size does not cancel:

$$\iint_\Omega u^2 dA_z = \iint_{\tilde{\Omega}} \frac{U(w)^2}{|f'(f^{-1}(w))|^2} dA_w.$$

Consequently a normalized eigenvalue problem cannot be transported by a conformal map without also transforming its weight. This leads to our next question.

To find a nonzero mode, remove the amplitude freedom

With homogeneous (zero) boundary data, minimizing a nonnegative quadratic energy alone usually gives $u = 0$. To select a *shape*, compare energy with squared size. On a finite interval define

$$\mathcal{Q}[u] = \int_a^b (pu'^2 + Vu^2) \, dx, \quad \mathcal{N}[u] = \int_a^b wu^2 \, dx, \quad \mathcal{R}[u] = \frac{\mathcal{Q}[u]}{\mathcal{N}[u]}. \quad (7.3.1)$$

For now take smooth real coefficients, $p, w > 0$, a nonzero real u , and $u(a) = u(b) = 0$. The potential V need not be positive.

For any real $c \neq 0$, both numerator and denominator acquire the factor c^2 : $\mathcal{R}[cu] = \mathcal{R}[u]$. Equivalently, minimize $\mathcal{Q}$ among shapes with $\mathcal{N} = 1$; rescaling u by $1/\sqrt{\mathcal{N}[u]}$ enforces this normalization.

$\mathcal{R}$ is the *Rayleigh quotient*. We will derive its equation directly by the quotient rule, without assuming a new optimization theorem.

Vary the quotient, not the normalization separately

The two variations are obtained by expanding squares:

$$\delta Q = 2 \int_a^b (pu'\eta' + Vu\eta) dx, \quad \delta N = 2 \int_a^b wu\eta dx.$$

At the candidate function let $\lambda = \mathcal{R}[u]$, a number. The quotient rule gives

$$\begin{aligned} \delta \mathcal{R} &= \frac{\mathcal{N}\delta Q - Q\delta \mathcal{N}}{\mathcal{N}^2} = \frac{\delta Q - \lambda\delta \mathcal{N}}{\mathcal{N}} \\ &= \frac{2}{\mathcal{N}} \left\{ [pu'\eta]_a^b + \int_a^b (-(pu')' + Vu - \lambda wu)\eta dx \right\}. \end{aligned}$$

The first variation of $Q - \lambda N$ is the same numerator when λ is held fixed. This explains the usual *Lagrange multiplier* form.

Here η may change the norm: the quotient automatically accounts for that change. There is no need to impose $\int wu\eta = 0$ in this calculation.

The equation is the one from Chapter 4

For zero endpoint variations, the boundary term vanishes. Stationarity gives

$$-(pu')' + Vu = \lambda wu. \quad (7.3.2)$$

Chapter 4 uses $(pu')' + qu = -\lambda wu$. The two conventions agree with

$$V = -q.$$

Multiplying (7.3.2) by u and integrating checks the eigenvalue:

$$-[pu'u]_a^b + \int_a^b (pu'^2 + Vu^2) dx = \lambda \int_a^b wu^2 dx, \quad \lambda = \mathcal{R}[u].$$

Other boundary conditions must make the relevant endpoint terms vanish. At a singular endpoint, both the energy and norm must be finite, and the limiting term must be checked, as in Chapters 4–5.

An arbitrary Robin condition generally requires a boundary-energy term. It is not obtained by silently discarding $[pu'\eta]_a^b$.

Why the lowest eigenvalue gives a lower bound

For the regular fixed-endpoint problem, use real eigenfunctions ϕ_j normalized by $\int w \phi_i \phi_j dx = \delta_{ij}$, with $\lambda_1 \leq \lambda_2 \leq \dots$. Integration by parts and their differential equations give

$$\int_a^b (p \phi_i' \phi_j' + V \phi_i \phi_j) dx = \int_a^b \lambda_i w \phi_i \phi_j dx = \lambda_i \delta_{ij}.$$

For a finite sum $u = \sum_j c_j \phi_j$, therefore,

$$\mathcal{R}[u] = \frac{\sum_j \lambda_j c_j^2}{\sum_j c_j^2} \geq \lambda_1. \quad (7.3.3)$$

It is a weighted average of eigenvalues. For this regular problem the same identity extends by the spectral expansion in the finite-energy class; we use that standard extension without proving it here. Thus **every admissible trial shape gives an upper bound on λ_1** . A boundary condition and a function class are part of this statement.

Higher modes are stationary, but are not all minima

Choose two normalized orthogonal modes and keep the norm exactly one:

$$u_\varepsilon = \frac{\phi_j + \varepsilon \phi_i}{\sqrt{1 + \varepsilon^2}}, \quad i \neq j.$$

Orthogonality makes the cross terms zero, so

$$\mathcal{R}[u_\varepsilon] = \frac{\lambda_j + \varepsilon^2 \lambda_i}{1 + \varepsilon^2}, \quad \mathcal{R}[u_\varepsilon] - \lambda_j = \frac{\varepsilon^2(\lambda_i - \lambda_j)}{1 + \varepsilon^2}.$$

The first derivative at zero vanishes. A lower-eigenvalue component decreases the quotient; a higher-eigenvalue component increases it.

To minimize for the next distinct mode, exclude lower modes by weighted orthogonality constraints. This explains why orthogonality is useful for *finding* eigenfunctions, not only for expanding functions after finding them. For repeated eigenvalues, different combinations in that eigenspace have the same quotient.

A beta integral estimates the first string eigenvalue

For $-u'' = \lambda u$ on $(0, 1)$ with $u(0) = u(1) = 0$, Chapter 4 gives $\lambda_1 = \pi^2$. Try $u = x(1 - x)$, which already fits both ends.

$$\mathcal{Q}[u] = \int_0^1 (1 - 2x)^2 dx = 1 - 2 + \frac{4}{3} = \frac{1}{3},$$

$$\mathcal{N}[u] = \int_0^1 x^2(1 - x)^2 dx = B(3, 3) = \frac{\Gamma(3)\Gamma(3)}{\Gamma(6)} = \frac{2!2!}{5!} = \frac{1}{30}.$$

Thus

$$\lambda_1 \leq \mathcal{R}[u] = 10, \quad \pi^2 \simeq 9.869604.$$

This is the *Rayleigh–Ritz idea*: optimize within a simpler family of functions. The beta function from Chapter 2 evaluates the trial norm; it is not an unrelated integration exercise. We will improve a disk trial using one adjustable coefficient.

The polynomial families share a variational form

For each family below, $V = 0$ and

$$\mathcal{R}[u] = \frac{\int_I p(x) u'(x)^2 dx}{\int_I w(x) u(x)^2 dx}, \quad -(pu')' = \lambda wu.$$

Mode	I	p	w	λ
P_n	$(-1, 1)$	$1 - x^2$	1	$n(n + 1)$
H_n	$\mathbb{R}$	e^{-x^2}	e^{-x^2}	$2n$
L_n	$(0, \infty)$	xe^{-x}	e^{-x}	n
T_n	$(-1, 1)$	$\sqrt{1 - x^2}$	$(1 - x^2)^{-1/2}$	n^2

Here $n = 0, 1, \dots$. These use the endpoint choices of Chapter 6, *not* fixed zero values. Polynomial modes and polynomial variations have finite integrals, and their terms $pu'\eta$ vanish at each limiting endpoint. The weights and endpoint behavior are essential parts of the problem.

Legendre illustrates the endpoint and zero-mode issues

Take $p = 1 - x^2$, $w = 1$. For smooth u, η on $[-1, 1]$, the boundary term $[(1 - x^2)u'\eta]_{-1}^1$ vanishes even though endpoint values are free. The equation is

$$-((1 - x^2)u')' = \lambda u \iff (1 - x^2)u'' - 2xu' + \lambda u = 0.$$

Chapter 6 identifies its regular modes as P_n , with $\lambda_n = n(n + 1)$. The constant $P_0 = 1$ has zero energy and is the lowest mode. For $P_1 = x$,

$$\mathcal{Q}[x] = \int_{-1}^1 (1 - x^2) dx = \frac{4}{3}, \quad \mathcal{N}[x] = \int_{-1}^1 x^2 dx = \frac{2}{3}, \quad \mathcal{R}[x] = 2.$$

To select this first nonconstant mode variationally, exclude the constant by $\int_{-1}^1 u dx = 0$. Minimizing without that constraint would simply return P_0 .

The normalized planar problem is Helmholtz, not Laplace

On a disk with fixed zero rim values, use

$$\mathcal{Q}[u] = \iint |\nabla u|^2 dA, \quad \mathcal{N}[u] = \iint u^2 dA, \quad \mathcal{R} = \mathcal{Q}/\mathcal{N}.$$

Here $\mathcal{Q} = 2\mathcal{D}$; this convention makes the quotient equal to λ . With $\eta = 0$ on the rim,

$$\delta(\mathcal{Q} - \lambda\mathcal{N}) = 2 \iint (\nabla u \cdot \nabla \eta - \lambda u \eta) dA = -2 \iint (\Delta u + \lambda u) \eta dA.$$

Hence

$$-\Delta u = \lambda u. \tag{7.4.1}$$

The normalization prevents $u = 0$. On a fixed disk the first eigenvalue is positive. For a membrane obeying $U_{tt} = c^2 \Delta U$, substitution of $U = u \cos \omega t$ gives $\lambda = \omega^2/c^2$. Thus the quotient selects a vibration frequency, not just a convenient constant.

Polar geometry supplies the Bessel weight

Use a unit disk. In polar coordinates,

$$|\nabla u|^2 = u_r^2 + \frac{u_\theta^2}{r^2}, \quad dA = r \, dr \, d\theta.$$

For a fixed integer angular order $m \geq 0$, set $u = R(r) \cos(m\theta)$. Then

$$u_r = R' \cos(m\theta), \quad u_\theta = -mR \sin(m\theta).$$

For $m \geq 1$, $\int_0^{2\pi} \cos^2(m\theta) \, d\theta = \int_0^{2\pi} \sin^2(m\theta) \, d\theta = \pi$. For $m = 0$, the cosine integral is 2π and the angular derivative is zero. The common angular factor cancels, leaving

$$\mathcal{R}_m[R] = \frac{\int_0^1 \left(rR'^2 + \frac{m^2}{r} R^2 \right) \, dr}{\int_0^1 rR^2 \, dr}. \quad (7.4.2)$$

The weight r comes from area; m^2/r comes from the angular gradient. The $m = 0$ potential term is absent. Sine modes for $m \geq 1$ give the same quotient.

Varying the radial energy gives Bessel's equation

Apply the first variation to half the numerator minus $\lambda/2$ times the denominator:

$$\begin{aligned} 0 &= \int_0^1 \left(rR'\eta' + \frac{m^2}{r} R\eta - \lambda rR\eta \right) dr \\ &= [rR'\eta]_0^1 + \int_0^1 \left(-(rR')' + \frac{m^2}{r} R - \lambda rR \right) \eta dr. \end{aligned}$$

Once the endpoint term vanishes, the radial equation is

$$(rR')' + \left(\lambda r - \frac{m^2}{r} \right) R = 0, \quad r^2 R'' + rR' + (\lambda r^2 - m^2)R = 0.$$

With $\lambda = k^2 > 0$, its solutions are

$$R(r) = AJ_m(kr) + BY_m(kr).$$

We have reached the same equation as the Frobenius calculation in Chapter 3. The admissible solution and the allowed k still depend on both endpoints.

At the center and the rim, check the actual terms

For a smooth disk mode, the behavior near the center is

$$\begin{aligned} m \geq 1 : \quad & R = O(r^m), \quad R' = O(r^{m-1}), \quad \eta = O(r^m), \\ m = 0 : \quad & R = R(0) + O(r^2), \quad R' = O(r), \quad \eta = O(1). \end{aligned}$$

Thus $rR'\eta \rightarrow 0$ in either case. The Y_m branch is excluded: for $m \geq 1$ it behaves like r^{-m} , and for $m = 0$ like $\log r$; its derivative energy diverges. At the fixed rim, $\eta(1) = 0$ and $R(1) = 0$. A nonzero regular solution therefore requires

$$J_m(k) = 0, \quad k = \alpha_{m,j}, \quad \lambda_{m,j} = \alpha_{m,j}^2, \quad j = 1, 2, \dots \quad (7.4.3)$$

Here $\alpha_{m,j}$ is the j th positive zero, as in Chapter 5. For a disk of radius L , rescaling $r = L\rho$ gives $\lambda_{m,j} = \alpha_{m,j}^2/L^2$. The endpoint conditions, not the differential equation alone, make the spectrum discrete.

A first disk trial requires only elementary integrals

For the axisymmetric sector $m = 0$, choose $R_0(r) = 1 - r^2$. It is smooth at the center and zero at the rim. Then

$$\mathcal{Q}_0[R_0] = \int_0^1 r(-2r)^2 dr = 4 \int_0^1 r^3 dr = 1,$$

$$\mathcal{N}_0[R_0] = \int_0^1 r(1 - r^2)^2 dr = \frac{1}{2} \int_0^1 (1 - t)^2 dt = \frac{1}{2} B(1, 3) = \frac{1}{6},$$

where $t = r^2$ and $r dr = dt/2$. Consequently

$$\alpha_{0,1}^2 \leq \mathcal{R}_0[R_0] = 6, \quad \alpha_{0,1}^2 \simeq 5.783186.$$

The minimum principle applies within the smooth axisymmetric sector of the disk; its lowest eigenvalue is $\alpha_{0,1}^2$. The subscript 0 on $\mathcal{Q}_0, \mathcal{N}_0, \mathcal{R}_0$ indicates $m = 0$; the trial curve is only an approximation.

Improve the shape with one adjustable coefficient

Take

$$R_a(r) = (1 - r^2)(1 + ar^2) = 1 + (a - 1)r^2 - ar^4, \quad a \in \mathbb{R}.$$

The boundary conditions are built in for every a . Differentiating gives $R'_a = 2(a - 1)r - 4ar^3$. Hence

$$\begin{aligned} \mathcal{Q}_0[R_a] &= \int_0^1 (4(a - 1)^2 r^3 - 16a(a - 1)r^5 + 16a^2 r^7) \, dr \\ &= (a - 1)^2 - \frac{8}{3}a(a - 1) + 2a^2 = 1 + \frac{2}{3}a + \frac{1}{3}a^2. \end{aligned}$$

For the norm, the substitution $t = r^2$ keeps the algebra shorter:

$$\begin{aligned} \mathcal{N}_0[R_a] &= \frac{1}{2} \int_0^1 (1 - t)^2 (1 + 2at + a^2 t^2) \, dt \\ &= \frac{1}{2} (B(1, 3) + 2aB(2, 3) + a^2 B(3, 3)) = \frac{1}{6} + \frac{a}{12} + \frac{a^2}{60}. \end{aligned}$$

The remaining minimization is ordinary calculus

The quotient is

$$\mathcal{R}_0[R_a] = 20 \frac{a^2 + 2a + 3}{a^2 + 5a + 10}, \quad a^2 + 5a + 10 = (a + 5/2)^2 + 15/4 > 0.$$

Differentiating gives

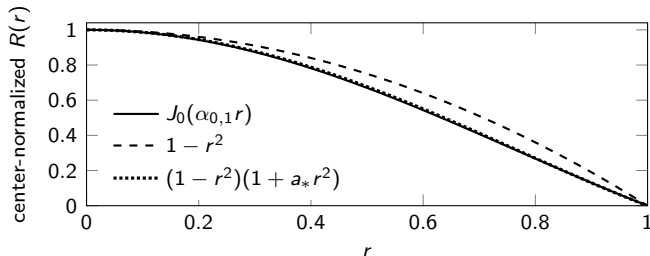
$$\frac{d\mathcal{R}_0}{da} = 20 \frac{(2a + 2)(a^2 + 5a + 10) - (a^2 + 2a + 3)(2a + 5)}{(a^2 + 5a + 10)^2} = 20 \frac{3a^2 + 14a + 5}{(a^2 + 5a + 10)^2}.$$

The two stationary parameters are $a_{\pm} = (-7 \pm \sqrt{34})/3$. At a stationary point $3a^2 + 14a + 5 = 0$, so $a^2 = -(14a + 5)/3$; substituting this into $\mathcal{R}_0$ collapses it to $\mathcal{R}_0 = 80(1 - 2a)/(a + 25)$, and the two roots give $(64 \mp 8\sqrt{34})/3$. Meanwhile the quotient tends to 20 as $|a| \rightarrow \infty$. Therefore the smaller value is the minimum in this trial family:

$$a_* = \frac{-7 + \sqrt{34}}{3}, \quad \alpha_{0,1}^2 \leq \frac{64 - 8\sqrt{34}}{3} \simeq 5.784128. \quad (7.4.4)$$

This remains an upper bound, not an exact eigenvalue.

What the approximation has, and has not, achieved



The upper bound improves from 6 to 5.784128, compared with 5.783186. The relative eigenvalue errors are about 3.75% and 0.0163%. The curves are scaled to $R(0) = 1$ for comparison; this is not unit weighted norm.

Adding admissible trial functions cannot increase the *best* upper bound, because the old trial set is still available. It does not make every added trial better.

A linear potential returns us to Airy functions

On $x \geq 0$, take $u(0) = 0$, decay at infinity, and finite norm and energy:

$$\mathcal{R}[u] = \frac{\int_0^\infty (u'^2 + xu^2) dx}{\int_0^\infty u^2 dx}.$$

As before, the endpoint terms must vanish. Its stationary equation is

$$-u'' + xu = \lambda u \iff u'' - (x - \lambda)u = 0.$$

Put $t = x - \lambda$. Since $dt/dx = 1$, this becomes $u_{tt} - tu = 0$, the Airy equation from Chapter 3. Thus

$$u(x) = A \operatorname{Ai}(x - \lambda) + B \operatorname{Bi}(x - \lambda).$$

Bi grows at positive infinity, so decay forces $B = 0$. The left endpoint gives

$$\operatorname{Ai}(-\lambda) = 0$$

for a nonzero solution. Airy zeros now select the permitted energy values, just as Bessel zeros selected the disk frequencies. Here x and λ are dimensionless.

A quadratic potential selects a Gaussian

On the whole real line, use real smooth rapidly decaying ψ with $\int \psi^2 dx = 1$. The dimensionless oscillator energy is

$$\mathcal{Q}[\psi] = \int_{-\infty}^{\infty} (\psi'^2 + x^2 \psi^2) dx.$$

Complete a square, retaining the integration-by-parts step:

$$\begin{aligned} \int_{-\infty}^{\infty} (\psi' + x\psi)^2 dx &= \mathcal{Q}[\psi] + \int_{-\infty}^{\infty} x(\psi^2)' dx \\ &= \mathcal{Q}[\psi] + [x\psi^2]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \psi^2 dx = \mathcal{Q}[\psi] - 1. \end{aligned}$$

Therefore $\mathcal{Q} \geq 1$. Equality requires

$$\psi' + x\psi = 0 \implies (e^{x^2/2}\psi)' = 0 \implies \psi = Ce^{-x^2/2}.$$

This proves the lowest energy and its minimizing shape, rather than only guessing them.

Gamma fixes the amplitude; Hermite gives the other modes

Normalization of the Gaussian uses Chapter 2:

$$1 = C^2 \int_{-\infty}^{\infty} e^{-x^2} dx = C^2 \Gamma(1/2) = C^2 \sqrt{\pi}.$$

Choose the positive sign: $\psi_0 = \pi^{-1/4} e^{-x^2/2}$. The stationary equation for a general normalized mode is $-\psi'' + x^2\psi = \lambda\psi$. To connect it to Chapter 6, write $\psi = e^{-x^2/2}h$ and differentiate:

$$\psi' = e^{-x^2/2}(h' - xh),$$

$$\psi'' = e^{-x^2/2}(h'' - 2xh' + (x^2 - 1)h).$$

Substituting and cancelling the nonzero Gaussian gives

$$-h'' + 2xh' + h = \lambda h \quad \Longleftrightarrow \quad h'' - 2xh' + (\lambda - 1)h = 0.$$

This is Hermite's equation when $\lambda = 2n + 1$ and $h = H_n$.

The two Hermite eigenvalue conventions agree

Chapter 6 gives the weighted polynomial norm

$$\int_{-\infty}^{\infty} e^{-x^2} H_n(x)^2 dx = 2^n n! \sqrt{\pi}.$$

Hence the normalized oscillator modes are

$$\psi_n(x) = \frac{e^{-x^2/2} H_n(x)}{\sqrt{2^n n! \sqrt{\pi}}}, \quad \lambda_n = 2n + 1, \quad n = 0, 1, \dots \quad (7.5.1)$$

The polynomial quotient earlier in this chapter has eigenvalues $2n$; the oscillator quotient has $2n + 1$. They act on different functions and use different weights:

$$H_n \text{ with weight } e^{-x^2} \longleftrightarrow e^{-x^2/2} H_n \text{ with weight } 1.$$

The $+1$ comes from differentiating the Gaussian, as shown on the preceding page. The higher modes are stationary, but the ground-state square completion already shows that only $n = 0$ attains the minimum energy 1.

A working procedure

- 1 **State the allowed functions.** Fix the boundary values or state which endpoint values are free. Require the needed energy and norm to be finite.
- 2 **Choose the functional.** Is the problem a fixed-boundary equilibrium, or a nonzero mode requiring normalization?
- 3 **Vary and integrate by parts.** Keep the boundary term until the allowed variations show what happens to it.
- 4 **Recognize and solve the equation.** Use the analytic functions, series, or special functions already developed.
- 5 **Interpret the result.** Prove a minimum by an energy difference or a spectral bound; otherwise call it stationary. Use a trial family when an exact solution is unavailable.

The essential distinction is between *deriving an equation, selecting its admissible solutions*, and *proving what they optimize*.

Practice: let one string endpoint be free

Question. For $\mathcal{R}[u] = \int_0^1 u'^2 dx / \int_0^1 u^2 dx$, fix $u(0) = 0$ but leave $u(1)$ free. Find the eigenvalues.

Solution. The boundary term gives $u'(1) = 0$, and the interior gives $-u'' = \lambda u$. Multiplying by u and integrating shows $\lambda \int u^2 = \int u'^2 > 0$: a zero derivative would make the admissible nonzero function a constant with $u(0) = 0$, which is impossible. Thus put $\lambda = k^2$, $k > 0$. Then

$$u = A \cos kx + B \sin kx, \quad u(0) = 0 \Rightarrow A = 0, \quad u'(1) = Bk \cos k = 0.$$

Since $B \neq 0$, $\cos k = 0$. Consequently

$$k = (n + \tfrac{1}{2})\pi, \quad \lambda_n = (n + \tfrac{1}{2})^2 \pi^2, \quad n = 0, 1, 2, \dots$$

Changing just one endpoint condition changes every allowed frequency.

Practice: the string spectrum returns to zeta

Question. For a string fixed at both ends of $(0, 1)$, find the sum of the reciprocals of its eigenvalues. Compare with the one-free-endpoint problem.

Fixed ends. Since $\lambda_n = (n\pi)^2$, $n \geq 1$,

$$\sum_{n=1}^{\infty} \frac{1}{\lambda_n} = \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\zeta(2)}{\pi^2} = \frac{1}{6}.$$

One free end. Since $(n + \frac{1}{2})^{-2} = 4(2n + 1)^{-2}$,

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{1}{(n + \frac{1}{2})^2 \pi^2} &= \frac{4}{\pi^2} \left(\sum_{j=1}^{\infty} \frac{1}{j^2} - \sum_{j=1}^{\infty} \frac{1}{(2j)^2} \right) \\ &= \frac{4}{\pi^2} \left(1 - \frac{1}{4} \right) \zeta(2) = \frac{1}{2}. \end{aligned}$$

The sums converge absolutely. Only the Chapter 2 value $\zeta(2) = \pi^2/6$ is used; no analytic continuation is required.